9. Compare Time and Space Complexity.

10. State the halting problem of TM.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Prove the equivalence of NFA's with or without ε – move. (11)

Or

12. (a) Construct a DFA equivalent to the regular expression $10+(0+11)0^*1$. (6)

(b) Write down the applications of finite Automata. (5)

UNIT II

13. Define the term 'Regular set". Prove that the class of regular set is closed under substitutions. (11)

Or

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Fourth Semester

Computer Science and Engineering

AUTOMATA LANGUAGES AND COMPUTATION

(2009 – 12 Common paper)

Time : Three hours | Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. List out the applications of automata theory.

2. What is Regular Language?

3. Define Induction principle.

4. What is the closure property of regular sets?

5. State the pumping lemma for Regular languages.

6. List out the properties of the CFL.

7. What is a turing machine?

8. What is the storage in FC?

14. Consider the CFG $G = (\{E, T, F\}, (+, *, (,), a\}, P, E)$

Where $P ::= \rightarrow E + T / T$

$T \rightarrow T^* F / F$

$F \rightarrow (E) / a$

Convert this Grammar to Chomsky normal form.
(1

UNIT III

15. Summarize the properties of Context-Fre Languages with suitable examples. (1

Or

16. State and explain push down Automata. Design push down Automata for the language $a^n b^n$ ov $\{a, b\}^+$. (1

UNIT IV

17. Design Turing Machine to recognize the langua $\{0^n, 1^n 0^n \; nl \geq 1\}$. (

00, 1100

Or

18. Describe the recursive and recursively enumera languages in detail. (

UNIT V

19. Illustrate the concept of NP-Complete problem with suitable examples. (11)

Or

20. Describe the time and space complexity of Turing Machine with suitable examples. (11)

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B.Tech. DEGREE EXAMINATION, MAY 2018.

Third Semester

Computer Science and Engineering

AUTOMATA LANGUAGES AND COMPUTATION

(2009 – 12 & 2013 – 14 onwards)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. Defines ε Closure Property.

2. Differentiate between Moore and Mealy machines.

3. When is a grammar ambiguous? Give example.

4. Define Greibach Normal Form.

5. What is Push down Automata?

6. List the closure properties of Context Free Languages.

7. What is a recursive language?

8. Compare total and partial recursive functions.

9. What is running time of Turing machine?

10. Which problems belong to Class P?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions.

11. Convert the following regular expression to DFA : $(0+1)^*(0+1)$.

Or

12. Prove that "If there is a language L, such that $L = L(A)$ for some DFA A, then there is a regular expression. R such that $L = L(R)$.

13. State and prove the pumping lemma of regular languages.

Or

14. Put the following grammar in CNF

$S \rightarrow AACD$

$A \rightarrow aAb|\varepsilon$

$C \rightarrow aC|a$

$D \rightarrow aDa|bDb|\varepsilon$

15. Convert PDA to CFG where PDA is given by $P(\{q_0, q_1\}, \{0, 1\}, \{X, Z_0\} \delta, q_o, Z_0\ q_1)$ Transition from 6 is defined by,

$$\delta(q_0, 0, Z_0) = (q_0, XZ_0)$$

$$\delta(q_0, 0, X) = (q_0, XX)$$

$$\delta(q_0, 1, X) = (q_1, \varepsilon)$$

$$\delta(q_1, 1, X) = (q_1, \varepsilon)$$

$$\delta(q_1, \varepsilon, X) = (q_1, \varepsilon)$$

$$\delta(q_1, \varepsilon, Z_0) = (q_1, \varepsilon)$$

Or

16. (a) Construct a PDA for the language $L = \{a^n b^{n+m} c^m / n \geq 1, m \geq 1\}$ (5)

(b) Show that $\{0^n 1^n 2^n / n \geq 1\}$ is not a Context Free Language. (6)

17. Construct Turing Machine for $L = \{0^n 1^n / n \geq 1\}$.

Or

18. Construct a Turing machine that can find the reverse of a binary string.

19. Show that the vertex cover problem is NP—Complete.

Or

20. Discuss in detail about NP-hardness and NP completeness. Also explain when a problem is NP-Hard and when it is NB-complete.

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B.Tech. DEGREE EXAMINATION, MAY 2015.

Fourth Semester

Computer Science and Engineering

AUTOMATA LANGUAGES AND COMPUTATION

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Construct NFA equivalent to the regular expression: $(0+1)01$.

2. Distinguish NFA and DFA.

3. State the relations, among regular expression, deterministic finite automaton, non deterministic finite automaton and finite automaton with epsilon transitions.

4. Find out the context free language $S \rightarrow aSb \mid aAb$; $A \rightarrow bAa \mid ba$;

5. Define sentential forms.

6. State the pumping lemma for Context free languages.

7. Does a NTM accept a language, accepted by TM (or DTM)? Justify.

8. Give the instantaneous description of Turing Machine.

9. When do you say a problem is undecidable? Mention any two undecidable problems.

10. Is the travelling salesman problem NP complete? If so, prove.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Consider the following $\in$-NFA.

does not

δ	$\in$	a	b
$\rightarrow p$	{r}	{q}	{p,r}
q	$\emptyset$	{p}	$\emptyset$
r*	{p,q}	{r}	{p}

(a) Compute $\in$-closure of each state. (3)

(b) Give the set of all strings of length 3 or less accepted by automaton. (2)

(c) Convert the given automaton to DFA. (6)

Or

12. (a) Prove: A language L is accepted by some ϵ-NFA if and only if L is accepted by some DFA. (6)

(b) Explain about Moore and Mealy machines. (5)

UNIT II

13. (a) Simplify the following grammar and find its equivalent in CNF : (5)

$$S \to AB \mid CA \quad B \to BC \mid AB \quad A \to a \quad C \to aB \mid b$$

(b) Find the GNF equivalent of the grammar $S \to AA \mid 0,\ A \to SS \mid 1$. (6)

Or

14. (a) Show that the grammar $S \to A \mid Sa \mid bSS \mid SSb \mid SbS$ is ambiguous. (5 + 6)

(b) Consider the productions $S \to aB \mid bA$; $A \to aS \mid bAA \mid a$; $B \to bS \mid aBB \mid b$. For the string aaabbabbba, find a left most derivation.

UNIT III

15. Design a PDA to accept palindromes over {a,b}. (11)

Or

16. (a) State the properties of CFL. (6)

(b) Show that CFL are closed under union operation but not under intersection. (5)

11 + 1111

2+5

UNIT IV

17. Design a Turing machine M that computes $x + y$ where x and y are positive integers. (11)

Or

18. Prove that the function $fadd(x, y) = x + y$ is a primitive recursive. (11)

UNIT V

19. Explain about the time and space complexity of Turing Machines. (11)

Or

20. Explain in detail about the NP complete problems with example. (11)

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016.

Fourth Semester

Computer Science and Engineering

AUTOMATA LANGUAGES AND COMPUTATION

(2009-2012 batches)

Time : Three hours Maximum : 75 marks

PART A – ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. Define DFA.

2. Differentiate Moore and Mealy machines.

3. List the closure properties of regular languages.

4. When a grammar is in Chomsky Normal Form?

5. Define PDA accepted by empty stack.

6. How can we prove that a given language is not a Context Free Language?

7. Define Recursive Enumerable Language.

8. Compare total and partial recursive functions.

9. What is running time of Turing machine?

10. Which problems belong to Class NP?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Convert the following regular expression to DFA: (11)
$(a+b)^*(a+b)$.

Or

12. Minimize the given DFA: (11)

Q/Σ	0	1
$\rightarrow A$	B	F
B	G	C
*C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

Also check whether the minimized DFA accepts the string "000101".

UNIT II

13. (a) Find the right most derivation and derivation tree for the string "abbaaba" using the following grammar : (5)

$S \to XaaX;\ X \to aX \mid bX \mid \varepsilon$.

(b) Eliminate ε-productions and unit productions from the following grammar : (6)

$E \to TA,\ A \to +TA \mid \varepsilon, T \to MB,\ B \to *MB \mid \varepsilon$,

$M \to \text{id}$.

Or

14. Put the grammar $S \to SS/0S1/01$ into a GNF.(11)

UNIT III

15. State and prove the pumping lemma of context free languages. (11)

Or

16. (a) Construct a PDA for the language $L = \{a^m b^n c^m / n \geq 1,\ m \geq 1\}$. (5)

(b) Show that $\{0^n 1^n 2^n / n \geq 1\}$ is not a Context Free Language. (6)

UNIT IV

17. Construct Turing Machine for binary language of palindrome strings. (11)

Or

18. Construct a Turing machine that can multiply 2 integers. (11)

UNIT V

19. (a) Explain the time and space complexity of Turing Machines. (5)

(b) Discuss in detail about problems of Classes P with example. (6)

Or

20. Write notes on NP-hardness and NP completeness. Also explain when a problem is NP Hard and when it is NP complete. (11)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Fourth Semester

Computer Science and Engineering

AUTOMATA LANGUAGES AND COMPUTATION

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Compare NFA and DFA.
2. Give the applications of finite Automata.
3. What is CNF?
4. What is a Context Free grammar?
5. What is meant by Push down Automata?
6. What is Bottom up Parsing?
7. What is a Turing Machine?
8. What is a Primitive Recursive function?

9. What is meant by Time complexity?

10. What is meant by NP class?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. Discuss about conversion of NFA to DFA. → Thir/ Prob

Or

12. Explain briefly Moore and Mealy Machines.

UNIT II

13. Discuss the properties of regular sets.

Or

14. Explain Greibach Normal Form.

UNIT III

15. Explain the properties of CFL.

Or

16. Discuss about decision algorithms.

UNIT IV

17. Discuss the working of various Turing Machines.

Or

18. Explain about Enumerable Languages.

UNIT V

19. Explain the Space complexity of Turing Machines.

Or

20. Discuss about NP Completeness.